

Energy Through Trophic Levels: Double-Entry Energy Accounting

Name: _____

Track what passes through each trophic level and what is lost. The two accounts must always reconcile.

The accounting rule. For every transfer,

$$\text{energy entering} = \text{energy passed through} + \text{energy wasted},$$

and likewise

$$100\% = \% \text{ passed through} + \% \text{ wasted}.$$

“Wasted” means energy that does not become biomass at the next trophic level. It may leave through respiration, movement, heat, waste, uneaten biomass, or incomplete digestion. Trophic transfer is often around 10%, although real values vary substantially.

1. One hectare of sunlight: open the energy ledger

The solar constant tells us how much Power (“Luminosity/Radiance”) the sun delivers per metre square hitting the Earth’s surface. It is given as

$$S = 1361 \text{ W m}^{-2}.$$

The “unit” of Power is the Watt (W) and the unit of length is the metre (m). This then says that if I could place a $1\text{m} \times 1\text{m}$ solar panel and it absorbed all the light (“photons”) about to hit the earth’s surface I could run $\frac{1361}{60} \approx 22$ 60W light bulbs instead. (A bit silly-why?).

Consider an idealised 10,000 m² field (one hectare) receiving this irradiance normally for one hour. Ignore atmosphere, clouds, day/night, and changing solar angle for this opening calculation.

a) Calculate the solar power incident on the field:

$$P = SA = (1361 \text{ W m}^{-2})(10,000 \text{ m}^2) = \text{_____ W}.$$

b) Calculate the solar energy arriving in one hour (= $60\text{s} \times 60\text{min}$). Whereas Power is measured in “Watts”, energy is measured in “Joules”.¹

Give your answer in Mega Joules MJ or Giga Joules, GJ. Mega, $M = 10^6$ Giga, 10^9

$$E = Pt = \text{_____}$$

Now follow this chain:

sunlight → grass → rabbit → fox.

The pass-through efficiencies are 1.8%, 12%, and 9% respectively.

Stage	Pass / %	Waste / %	Energy in / MJ	Passed / MJ	Wasted here / MJ	Cumulative waste / MJ
Sun → grass	1.8					
Grass → rabbit	12					
Rabbit → fox	9					

c) At each row, check that **passed + wasted here = energy in**.

d) What percentage of the original solar energy finally passes through into fox biomass? _____%

e) What percentage has therefore been cumulatively wasted? _____%

f) Double-entry cross-check:

$$\text{final fox energy} + \text{cumulative wasted energy} = \text{_____ MJ}.$$

Does this equal the original solar input? _____

¹One Scotsman the other a Frenchman. The butterfly flapping sources of the industrial revolution and colonisation, peut-etre?

2. Three more trophic ledgers

For each pathway, calculate both the pass-through percentage and the wasted percentage at every step. Then calculate final pass-through and cumulative waste.

A. Pond chain 15,000 kJ sunlight → algae (2.5%) → zooplankton (18%) → small fish (14%) → pike (7%).

Transfer	Pass / %	Waste / %	Passed energy / kJ	Wasted here / kJ
Sun → algae	2.5			
Algae → zooplankton	18			
Zooplankton → fish	14			
Fish → pike	7			

Final pass-through = _____%Cumulative waste = _____%.

B. Crop eaten directly 10,000 kJ sunlight → crop biomass (2.2%) → human biomass (25%).

Crop pass / waste = 2.2% / _____%Human pass / waste = 25% / _____%.Energy in crop = _____ kJEnergy reaching human = _____ kJ.Final pass-through = _____%Cumulative waste = _____%.

C. Woodland chain 30,000 kJ sunlight → leaves (1.2%) → caterpillar (15%) → songbird (11%) → hawk (8%).

Complete the energy cascade on scrap paper or below. At every transfer record both pass and waste.

Final hawk energy = _____ kJFinal pass-through = _____%Cumulative waste = _____%Check: final energy + total wasted = _____ kJ.

Discover the Multiplicative RuleName: _____

3. Pass-through percentages: what operation combines them?

For the meadow chain, the stage pass-through efficiencies were 1.8%, 12%, and 9%.

- a) Add the three pass-through percentages. What do you get? _____%. Does that agree with the final pass-through? _____
- b) Write them as decimal fractions: _____, _____, _____.
- c) Find an operation on those three decimals that reproduces your final pass-through fraction:
- d) Complete the rule:

$\eta_{\text{total}} =$
- e) Why does this operation make physical sense? Complete the sentence:
Each new percentage acts on _____, rather than on the original input.

4. Waste gives an independent route to the same answer

For any whole pathway,

$$\% \text{ cumulative waste} = 100\% - \% \text{ final pass-through.}$$

This is **not** found by simply adding the stage waste percentages.

- a) For a two-step chain with 20% then 30% pass-through, calculate the final pass-through. _____%
- b) Hence find cumulative waste. _____%
- c) The stage waste percentages are 80% and 70%. Why is $80\% + 70\% = 150\%$ not the cumulative waste?
- d) Starting with 1000 kJ, show the two waste entries and verify:

$$\text{final passed energy} + \text{waste at stage 1} + \text{waste at stage 2} = 1000 \text{ kJ.}$$

5. Direct calculation: close both accounts

Complete the final pass-through and cumulative waste without calculating every intermediate energy.

Pathway	Stage pass-through efficiencies	Final pass / %	Cumulative waste / %
Grass → vole → owl	8%, 13%		
Phytoplankton → krill → penguin → seal	16%, 11%, 9%		
Maize → insect → chicken → human	7%, 18%, 12%		

6. Reverse the ledger

A chain has three successive pass-through efficiencies. The first is 10%, the second is 15%, and the overall pass-through is 0.12%.

- a) Find the third pass-through efficiency. _____%
- b) Find its corresponding waste percentage. _____%
- c) Find the cumulative waste percentage for the entire chain. _____%

7. Interpretation

- a) Which pathway in Section 2 has the greatest final pass-through? _____
- b) Why does shortening a food chain often give a large energetic advantage?
- c) Explain why “90% is wasted at each trophic level” does not mean that 90% of the original energy is lost again at every level.

Close the books. Complete both statements without looking back.
Successive pass-through efficiencies combine by _____.
Final pass-through % + cumulative waste % = _____.